

Relational Model

3NF relations with Primary and Foreign Keys identified:

R1

eBook(eBookID, bookTitle, bookDescription, releaseYear, availableStatus, authorID, categoryID, publisherID, branchID, formatID, librarianID)

Primary Key: eBookID

Foreign Key: authorID references R7(authorID)

Foreign Key: categoryID references R8(categoryID)

Foreign Key: publisherID references R6(publisherID)

Foreign Key: branchID references R5(branchID)

Foreign Key: formatID references R10(formatID)

Foreign Key: librarianID references R9(librarianID)

R2

loans (loanID, loanDate, returnDate, eBookID, memberID)

Primary Key: loanID

Foreign Key: eBookID references R1(eBookID)

Foreign Key: memberID references R3(memberID)

R3

members (memberID, firstName, lastName, email, phoneNumber, accountStatus)

Primary Key: memberID

R4

reservations (reservationID, reservationDate, status, eBookID, memberID)

Primary Key: reservationID

Foreign Key: eBookID references R1(eBookID)

Foreign Key: memberID references R3(memberID)

R5

branch (branchID,branchName,branchLocation)

Primary Key: branchID

R6

publisher (publisherID,publisherName)

Primary Key: publisherID

R7

author (authorID, authFirstName, authLastName)

Primary Key: authorID

R8

category (categoryID,categoryName)

Primary Key: categoryID

R9

librarian (librarianID,librarianFirstName,librarianLastName)

Primary Key: librarianID

R10

format (formatID,formatType)

Primary Key: formatID